Two corrections to the diagrams for Sz(8)

Human Compression Project

Abstract

The subgroup descriptions in the rank-two Suzuki classification determine the local residue orders without any graph computation. They also disprove two quasi-primitivity labels under the definition printed in the source. These corrections concern four numbered diagrams. The girths and diameters of the twenty-nine diagrams require a separate finite verification.

1 Residue orders and the printed orientation

Let G be a finite group, and let the two parabolics of a coset geometry be G_0, G_1, with intersection J. Its type-i elements are the left cosets of G_i, so their number is $n_i = [G : G_i]$. The type-i elements incident with the fixed element G_j, for $j \neq i$, are precisely

$$\{gG_i : g \in G_j\}.$$

Two representatives give the same coset precisely when they differ by an element of J. The number of these elements is therefore $[G_j : J]$. The source defines the order s_i at its type-i vertex as this residue cardinality minus one [1, p. 100]. Thus

$$(s_0, s_1) = ([G_1 : J] - 1, [G_0 : J] - 1). \tag{1}$$

The order at a diagram vertex is not the valency of that vertex in the incidence graph: that valency counts elements of the other type.

Proposition 1. *In source diagrams 25 and 26 [1, p. 118], the displayed orders* $(12, 4)$ *must be replaced by* $(4, 12)$*, with the printed parabolic orientation retained.*

Proof. The left parabolic is $C_{13} \rtimes C_4$, of order 52, and the right parabolic is $C_5 \rtimes C_4$, of order 20. Their common subgroup has order 4. Formula (1) gives $20/4-1 = 4$ on the left and $52/4-1 = 12$ on the right. The printed element counts $(560, 1456)$ independently fix this orientation, since $|\operatorname{Sz}(8)| = 29120$ and

$$29120/52 = 560, \qquad 29120/20 = 1456.$$

The graph edge labels and the two parabolic labels are not changed by this correction. □

2 The quasi-primitivity labels

The source calls a subgroup quasi-maximal when there is a unique chain of intermediate subgroups between it and the ambient group, and calls the geometry quasi-primitive when both parabolics have this property [1, p. 99]. The following argument supplies three different saturated chains for the parabolic in question.

Use the root coordinates already established in the normalizer treatment:

$$P = \{u(a,b) : a,b \in \mathbb{F}_8\}, \quad u(a,b)u(c,d) = u(a+c, b+d+ac^4).$$

Let $Z = \{u(0,b) : b \in \mathbb{F}_8\}$, and let $w = u(1,0)$. Then P has order 64, Z has order 8, and

$$R = \{u(a,b) : a \in \mathbb{F}_2,\ b \in \mathbb{F}_8\} \cong C_4 \times C_2^2$$

has order 16. These identifications, and the normalizer $N_G(\langle w \rangle) = R$, are acquired from the accepted root calculation.

Proposition 2. *The $C_4 \times C_2$ parabolic in source diagrams 27 and 29 is not quasi-maximal in* Sz(8)*. Both printed QPRI labels must be removed.*

Proof. Write $H \cong C_4 \times C_2$ for that parabolic. Its intersection with the other parabolic is a cyclic subgroup J of order four. All such cyclic subgroups are conjugate, so conjugate the configuration until $J = \langle w \rangle$. Since H is abelian, it normalizes J, and hence $H \leq R$. Its order is 8, so $[R : H] = 2$.

The first-coordinate map $P \to (\mathbb{F}_8, +)$ is a surjective homomorphism with kernel Z. The subgroup R is its inverse image of $\mathbb{F}_2$. Consequently $R \lhd P$ and

$$P/R \cong (\mathbb{F}_8/\mathbb{F}_2, +) \cong C_2^2.$$

The latter group has three different subgroups of order two. Their inverse images are three different subgroups P_1, P_2, P_3 of order 32. For the point stabilizer $B = P \rtimes C_7$, we obtain the three chains

$$H < R < P_i < P < B < G, \qquad i = 1,2,3.$$

The first three inclusions each have index two, $[B : P] = 7$, and B is a maximal subgroup of G by the acquired Suzuki subgroup theorem. Thus each chain is saturated and the chains are distinct. This disproves uniqueness even if "chain" is read as "maximal chain". One parabolic failing the definition suffices to remove QPRI from the geometry. □

The proof uses the source's own definition and the already acquired root and maximal-subgroup results. It does not require a subgroup census. The local double-transitivity label on diagram 29 is independent of this correction and is retained by the general classification theorem.

References

[1] D. Leemans, *The rank 2 geometries of the simple Suzuki groups* Sz(q), Beiträge zur Algebra und Geometrie **39** (1998), 97–120.